

7. Module-by-Module Demo Matrix

Module	Demo workflow	Saved result behavior
Platform and Projects	Create or open local folder projects, register project.json, upload project files, and continue previous live work.	Autosaves project history, result metadata, and live view state where a live result is generated.
Dashboard	Browse uploaded files, generated results, exports, project history, and open saved results back into live modules.	Autosaves project history, result metadata, and live view state where a live result is generated.
Log Visualization	Upload LAS, inspect well metadata, select curves, render multi-track depth plots, histogram, and crossplot tabs.	Autosaves project history, result metadata, and live view state where a live result is generated.
Missing Log Prediction	Analyze missing curve intervals, select target/features/model, predict gaps, save prediction JSON and live result state.	Autosaves project history, result metadata, and live view state where a live result is generated.
AI Facies Classification	Cluster or classify facies from LAS/table data, render facies tracks and preview classified rows.	Autosaves project history, result metadata, and live view state where a live result is generated.
AI Formation Tops	Detect formation tops from log response changes or uploaded tops/manual tops, then export results.	Autosaves project history, result metadata, and live view state where a live result is generated.
AI Parameter Prediction	Calculate Vsh, porosity, water saturation, lithology, confidence, and preview rows from active LAS.	Autosaves project history, result metadata, and live view state where a live result is generated.
AI Uncertainty	Generate P10/P50/P90 envelopes for porosity and saturation from prediction-ready LAS sessions.	Autosaves project history, result metadata, and live view state where a live result is generated.
Auto Splicer	Validate multiple LAS intervals, splice into a merged LAS, preview output tracks, and download LAS/provenance.	Autosaves project history, result metadata, and live view state where a live result is generated.
Crossplot	Create interactive X/Y scatter plots with optional color-by curve and depth filters.	Autosaves project history, result metadata, and live view state where a live result is generated.
Histogram	Create distribution plots, KDE, statistics, percentiles, quality score, and downloadable figure views.	Autosaves project history, result metadata, and live view state where a live result is generated.
Seismic Frequency Enhancer	Upload SEG-Y/SEG-Y/NPY, choose view/workflow/frequency controls, generate enhanced seismic panels.	Autosaves project history, result metadata, and live view state where a live result is generated.
Production Intelligence	Analyze production CSV/demo data, show optimization candidates, KPIs, and operating recommendations.	Autosaves project history, result metadata, and live view state where a live result is generated.
CCUS Screening	Map LAS curves, calculate reservoir-seal candidates, rank zones, show log panels, and export screening tables.	Autosaves project history, result metadata, and live view state where a live result is generated.
Geothermal Screening	Analyze LAS temperature and reservoir quality, detect hot zones, heat-flow map, and export results.	Autosaves project history, result metadata, and live view state where a live result is generated.
Digitizer SLM/GPT and OCR	Prepare document/image digitization workflows for extraction, review, assistant interaction, and export.	Autosaves project history, result metadata, and live view state where a live result is generated.
Reports and Settings	Provide reporting/export surfaces and workspace preferences without changing module result state.	Autosaves project history, result metadata, and live view state where a live result is generated.

8. Result Restore and Download Behavior

User action	Expected behavior
Generate a module result	Result payload writes to results/*.json and current live view writes to project.json.
Open another project	That project loads only its own saved files, result list, and live views.
Return to previous project	Previous live graphs and tables appear again after folder-backed hydration.
Click Dashboard Result List row	The app reads exact saved result JSON and opens the correct module route.
Download export	Export file is copied to project results exports folder and recorded in Export List.
Browser storage quota	Only compact metadata is cached; large result data remains inside the project folder.

Production acceptance checklist

- Open project folder permission is requested when needed by the browser File System Access API.
- project.json remains the source of truth for live module state and result history.
- Saved result payloads are kept as JSON for reproducible live views.
- Large graph payloads are not pushed into localStorage.
- All UI pages hydrate from the project folder before rendering live module state.

9. Demo Script for Stakeholders

Step	Presenter action	Talking point
1	Create or open a project folder	Drake AI runs as a project-first workspace.
2	Upload LAS/CSV/SEG-Y files	Inputs are copied to structured project storage.
3	Open Log Visualization and run plot	Multi-track graphs are generated from the active LAS.
4	Run Crossplot or Histogram	Result visualizations are saved and appear in Dashboard Result List.
5	Switch project and reopen the first project	Live graphs/data are project-specific and restored from project.json.
6	Click a saved result row	The exact historical result is reopened in the live module page.
7	Export/download outputs	Artifacts are tracked in Export List and stored inside the project folder.

End of demo PDF. This document is generated from the current Drake AI workspace and is intended as a downloadable stakeholder walkthrough.